

IB Bot

Production Week Plan

February 7 - 13, 2026

This plan covers 7 days of focused work to bring the IB Bot trading system from its current state to production readiness. Each day is designed for ~1 hour of your time with Cursor agents doing the heavy lifting.

Goal: Full backtesting suite, portfolio optimizer, and live trading system ready for deployment on a VM with Interactive Brokers.

Current State (Day 1 Complete)

Strategies:	22 total, 20 enabled, ALL 20 backtested LIVE
Price Cache:	2,424 IB tickers + 1,743 YF tickers
SEC EDGAR:	3 fund filings fully cached (Burry 32, Ackman 41, Marks 103)
QuiverQuant:	LIVE - API key active, 17 strategies using fresh signals
Unified Runner:	run_all_backtests.py - single source of truth
Data Quality:	Insider Purchases anomaly RESOLVED (296% -> 21% CAGR)

Critical Constraints

- 1 hour/day of your time - Cursor agents handle all coding
- Real money at stake (5% NW) - safety checks are non-negotiable
- VM migration needed for IB Gateway - requires your manual setup

Week Overview

- [x] Day 1 (Sat): Backtest Foundation
- [] Day 2 (Sun): Stats Validation & Regression Tests
- [] Day 3 (Mon): UI: Real Data, No Synthetic Charts
- [] Day 4 (Tue): Portfolio Optimizer
- [] Day 5 (Wed): Paper Trading Automation
- [] Day 6 (Thu): Live Trading Safety & IB Integration
- [] Day 7 (Fri): Integration Testing & Go-Live

DAY 1 (Saturday) - BACKTEST FOUNDATION

STATUS: COMPLETED

Deliverables

- [x] Unified backtest runner (run_all_backtests.py)
- [x] Strategy registry - canonical list of all 22 strategies
- [x] Data coverage report with date ranges + quality flags
- [x] Extended metrics: Sharpe, Sortino, Calmar, DD duration, profit factor
- [x] Cache fallback - works offline when QuiverQuant is down
- [x] ALL 20 enabled strategies backtested with fresh live data
- [x] Data quality flags (Insider Purchases anomaly RESOLVED)
- [x] SEC EDGAR cache fully populated (Oaktree 103 filings)
- [x] Fixed sec13F fallback loop (no more Quiver premium errors)
- [x] Howard Marks backtest working (was hanging, now 0.4s)

All 20 Strategies - Fresh Live Data (sorted by CAGR)

Strategy	CAGR	Shrp	Sort	MaxDD	Alpha	Beta	WinR
Michael Burry	44.3%	1.31	2.03	-39.7%	57.1%	0.05	53.4%
Congress Long-Short	29.1%	0.91	1.28	-35.4%	26.9%	0.03	53.7%
Transport. & Infra. Cmte	28.5%	1.15	1.64	-31.4%	21.9%	0.03	54.8%
U.S. House Long-Short	26.2%	1.06	1.64	-32.2%	20.3%	0.04	54.4%
Energy & Commerce Cmte	23.7%	0.98	1.44	-33.8%	18.6%	0.02	53.2%
Sector Weighted DC Insider	22.8%	1.22	1.72	-18.0%	17.7%	0.01	55.3%
Congress Buys	21.2%	0.96	1.34	-29.1%	17.1%	0.03	55.4%
Insider Purchases	21.0%	0.64	1.09	-39.5%	17.5%	0.06	50.8%
Dan Meuser	19.2%	0.68	0.92	-37.1%	19.1%	0.01	54.6%
Top Lobbying Spenders	17.6%	0.76	1.02	-30.9%	18.9%	0.02	53.5%
Josh Gottheimer	16.1%	0.76	0.98	-32.6%	12.3%	0.01	54.0%
Bill Ackman	15.2%	0.62	0.77	-46.9%	17.2%	0.04	55.0%
Homeland Sec. Cmte	14.3%	0.66	0.89	-29.0%	9.7%	0.03	52.8%
Top Gov Contracts	12.6%	0.48	0.62	-41.0%	15.5%	0.00	52.8%
Congress Sells	11.6%	0.45	0.62	-27.0%	7.1%	0.02	53.6%
Sheldon Whitehouse	11.2%	0.63	0.78	-28.9%	10.9%	0.01	55.1%
Donald Beyer	9.4%	0.40	0.54	-27.0%	12.6%	-0.02	53.2%
Lobbying Spending Growth	9.0%	0.27	0.36	-43.5%	11.0%	0.00	52.4%
Howard Marks	4.7%	0.16	0.23	-21.0%	9.5%	0.01	53.3%
Nancy Pelosi	4.5%	0.12	0.17	-42.0%	6.1%	0.04	52.6%

DAY 2 (Sunday) - STATS VALIDATION & REGRESSION TESTS

Problem

Need full confidence in backtest numbers. Some cached equity curves may have drift from the live engine. No regression tests to catch future breakage.

Cursor Agents Will

- Add missing stats to the unified runner: rolling Sharpe (12mo window), drawdown recovery time, skewness, kurtosis
- Cross-validate: compare cached plot_data.json CAGRs against fresh RebalancingBacktestEngine runs for all 20 strategies
- Build regression test suite: snapshot current results, assert future runs produce identical numbers (determinism check)
- Insider Purchases anomaly RESOLVED (Day 1): was corrupted cached data with 400%+ single-period jumps. Fresh data shows 21.0% CAGR.
- Create a BacktestReport class that outputs structured JSON with all stats, consumable by the UI

- [] Review stat output for your best 5 strategies
- [] Flag anything that looks wrong vs your expectations
- [] Approve the regression test baselines

Comprehensive stats for every strategy with regression tests that lock in known-good results. One command: `python -m pytest tests/test_backtest_regression.py`

DAY 3 (Monday) - UI: REAL DATA, NO SYNTHETIC CHARTS

Problem

UI shows synthetic/fake charts for some strategies. Missing strategies in the catalog. Charts don't use real backtest data from the unified runner.

Cursor Agents Will

- Trace the chart data pipeline: API endpoint -> plot_data.json -> frontend Chart.js component
- Replace any synthetic chart generation with real data from .cache/all_results_combined.json
- Add all 22 strategies to the strategy catalog UI
- Add 'last updated' timestamp and 'data source' badge to every chart (live vs cached)
- Ensure /api/plot_data serves real backtest curves from the unified runner
- Fix any discrepancies between plot_data.json and latest_backtest_results.json

- [] Open the UI (docker-compose up)
- [] Check every strategy page - verify charts match Day 2 numbers
- [] Flag any remaining synthetic or stale data

UI showing only real backtest data. Every chart has a timestamp and source badge. All 22 strategies visible in the catalog.

DAY 4 (Tuesday) - PORTFOLIO OPTIMIZER

Problem

Need multi-strategy portfolio construction with weight optimization. Current `portfolio_math.py` and `portfolio_backtest.py` exist but need verification and enhancement.

Cursor Agents Will

- Verify `portfolio_math.py` `nav_blend` produces correct blended NAV curves
- Add portfolio optimization methods: equal-weight, inverse-volatility, risk-parity, and max-Sharpe (mean-variance)
- Build portfolio comparison tool: same strategies, different weight schemes, side-by-side results
- Add constraints: max weight per strategy (e.g., 30%), min diversification, max correlation between strategies
- Connect portfolio backtester to the allocations UI page
- Create a recommended portfolio based on Day 2 validated stats

- [] Create 2-3 portfolio configs with your preferred strategies
- [] Run the optimizer - review suggested weights
- [] Compare blended performance vs individual strategies
- [] Select your V1 portfolio allocation

Working portfolio optimizer: suggest weights, backtest blended result, compare schemes. V1 allocation locked in.

DAY 5 (Wednesday) - PAPER TRADING AUTOMATION

Problem

Paper trading exists (dual implementation: SQLite legacy + PostgreSQL modern) but isn't automated. No scheduled rebalancing. Need to validate the full signal -> rebalance -> fill -> P&L cycle.

Cursor Agents Will

- Consolidate to PostgreSQL paper trading only (deprecate SQLite)
- Build automated rebalancing via Celery beat: check signals on schedule, execute paper rebalances
- Add daily P&L snapshots and portfolio tracking
- Create historical simulation mode: fast-forward through historical data to validate full cycle
- Add paper trading dashboard data to the API
- Write integration test: create portfolio -> generate signals -> rebalance -> verify positions

- [] Run the historical simulation
- [] Verify paper trading results are consistent with backtests
- [] Review automated rebalancing schedule

Automated paper trading that rebalances on schedule with P&L tracking. Historical simulation validates the full cycle.

DAY 6 (Thursday) - LIVE TRADING SAFETY & IB INTEGRATION

Problem

Live trading infrastructure exists but execution is intentionally disabled. Need comprehensive safety checks before enabling with real money. VM deployment preparation needed.

Cursor Agents Will

- Review and harden all safety checks: position size limits, max daily loss, max orders/day, spread checks, price validation
- Add pre-trade checklist: account balance, position limits, correlation, market hours verification
- Build 'dry run' mode: log exactly what orders would be placed without executing
- Create VM deployment script: docker-compose with IB Gateway, env setup, health checks, auto-restart
- Add alerting: Telegram/email notifications for trades, errors, daily P&L
- Document IB Gateway setup steps for VM (manual steps you'll need to do)
- Create kill-switch: one command to halt all trading and liquidate

- ☐ Review every safety check - set your personal risk limits
- ☐ Set max position size (e.g., 10% of portfolio per position)
- ☐ Set max daily loss limit (e.g., 2% of portfolio)
- ☐ Begin VM setup if available

Live trading system with comprehensive safety rails. VM deployment script ready. Kill-switch operational. Risk limits configured.

DAY 7 (Friday) - INTEGRATION TESTING & GO-LIVE

Problem

No full-stack testing. Need confidence that all components work together before deploying with real money.

Cursor Agents Will

- Write end-to-end integration tests: API -> backtest -> results -> UI
- Write paper trading integration test: create portfolio -> signals -> rebalance -> verify positions
- Test full Docker stack boots cleanly (db + redis + api + worker + beat + web)
- Create production runbook: startup, monitoring, troubleshooting, rollback
- Final bug-fix pass on anything found during the week
- Create go-live checklist with manual verification steps

- [] Run the full test suite
- [] Walk through the go-live checklist
- [] Make the go/no-go decision
- [] If go: deploy to VM, connect IB Gateway, start paper trading

Tested system with production runbook. Go-live checklist completed. System deployed and running in paper mode.

What 'Done' Looks Like - Friday Night

Backtesting	All strategies, single engine, consistent data, full stats, regression tests
Portfolio Optimizer	Multi-strategy blends with weight optimization (4 methods)
Paper Trading	Automated rebalancing on schedule, daily P&L tracking
Live Trading	Safety-hardened, dry-run tested, IB Gateway ready on VM
UI Dashboard	Real charts only, all strategies, portfolio allocations
Testing	Integration tests passing, production runbook written

Risks & Mitigations

Risk: QuiverQuant stays down

Mitigation: Use cached equity curves + SEC EDGAR. 17 strategies still have cached data. If restored mid-week, re-run fresh backtests immediately.

Risk: IB Gateway VM setup

Mitigation: Requires your manual setup (agents can't SSH). Day 6 prep maximizes automation. Fallback: run locally for first week.

Risk: Time overrun

Mitigation: If something breaks unexpectedly, cut Day 5 (paper trading automation) - it's least critical for go-live. Can run paper trades manually.

Risk: Data quality issues

Mitigation: Insider Purchases 296% CAGR flagged. Regression tests on Day 2 will catch any other anomalies. Don't trade any strategy with unflagged data.

Quick Reference Commands

```
# Coverage report
python run_all_backtests.py --report-only

# Run all backtests (cache fallback)
python run_all_backtests.py --output results.json

# Run SEC EDGAR strategies only (no API key needed)
python run_all_backtests.py --sec-edgar-only

# Run specific strategies
python run_all_backtests.py --strategies "Michael Burry,Congress Buys"

# Core strategies only
python run_all_backtests.py --category core

# Start full stack
docker-compose up -d
```